

Then f is entire, and it is easy to see that

$$|f(z)| \leq e^{|z|^{1/2}},$$

and the order of growth of f is $1/2$. Moreover, $f(z)$ vanishes when $z_n = ((n + 1/2)\pi)^2$, while $\sum_n 1/|z_n|^s < \infty$ exactly when $s > 1/2$.

A natural question is whether or not, given any sequence of complex numbers $z_1, z_2, \dots$, there exists an entire function f with zeros precisely at the points of this sequence. A necessary condition is that $z_1, z_2, \dots$ do not accumulate, in other words we must have

$$\lim_{k \rightarrow \infty} |z_k| = \infty,$$

otherwise f would vanish identically by Theorem 4.8 in Chapter 2. Weierstrass proved that this condition is also sufficient by explicitly constructing a function with these prescribed zeros. A first guess is of course the product

$$(z - z_1)(z - z_2) \cdots,$$

which provides a solution in the special case when the sequence of zeros is finite. In general, Weierstrass showed how to insert factors in this product so that the convergence is guaranteed, yet no new zeros are introduced.

Before coming to the general construction, we review infinite products and study a basic example.

3 Infinite products

3.1 Generalities

Given a sequence $\{a_n\}_{n=1}^{\infty}$ of complex numbers, we say that the product

$$\prod_{n=1}^{\infty} (1 + a_n)$$

converges if the limit

$$\lim_{N \rightarrow \infty} \prod_{n=1}^N (1 + a_n)$$

of the partial products exists.

A useful necessary condition that guarantees the existence of a product is contained in the following proposition.

Proposition 3.1 *If $\sum |a_n| < \infty$, then the product $\prod_{n=1}^{\infty} (1 + a_n)$ converges. Moreover, the product converges to 0 if and only if one of its factors is 0.*

This is simply Proposition 1.9 of Chapter 8 in Book I. We repeat the proof here.

Proof. If $\sum |a_n|$ converges, then for all large n we must have $|a_n| < 1/2$. Disregarding if necessary finitely many terms, we may assume that this inequality holds for all n . In particular, we can define $\log(1 + a_n)$ by the usual power series (see (6) in Chapter 3), and this logarithm satisfies the property that $1 + z = e^{\log(1+z)}$ whenever $|z| < 1$. Hence we may write the partial products as follows:

$$\prod_{n=1}^N (1 + a_n) = \prod_{n=1}^N e^{\log(1+a_n)} = e^{B_N},$$

where $B_N = \sum_{n=1}^N b_n$ with $b_n = \log(1 + a_n)$. By the power series expansion we see that $|\log(1 + z)| \leq 2|z|$, if $|z| < 1/2$. Hence $|b_n| \leq 2|a_n|$, so B_N converges as $N \rightarrow \infty$ to a complex number, say B . Since the exponential function is continuous, we conclude that e^{B_N} converges to e^B as $N \rightarrow \infty$, proving the first assertion of the proposition. Observe also that if $1 + a_n \neq 0$ for all n , then the product converges to a non-zero limit since it is expressed as e^B .

More generally, we can consider products of holomorphic functions.

Proposition 3.2 *Suppose $\{F_n\}$ is a sequence of holomorphic functions on the open set Ω . If there exist constants $c_n > 0$ such that*

$$\sum c_n < \infty \quad \text{and} \quad |F_n(z) - 1| \leq c_n \quad \text{for all } z \in \Omega,$$

then:

- (i) *The product $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$.*
- (ii) *If $F_n(z)$ does not vanish for any n , then*

$$\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F'_n(z)}{F_n(z)}.$$

Proof. To prove the first statement, note that for each z we may argue as in the previous proposition if we write $F_n(z) = 1 + a_n(z)$, with

$|a_n(z)| \leq c_n$. Then, we observe that the estimates are actually uniform in z because the c_n 's are constants. It follows that the product converges uniformly to a holomorphic function, which we denote by $F(z)$.

To establish the second part of the theorem, suppose that K is a compact subset of Ω , and let

$$G_N(z) = \prod_{n=1}^N F_n(z).$$

We have just proved that $G_N \rightarrow F$ uniformly in Ω , so by Theorem 5.3 in Chapter 2, the sequence $\{G'_N\}$ converges uniformly to F' in K . Since G_N is uniformly bounded from below on K , we conclude that $G'_N/G_N \rightarrow F'/F$ uniformly on K , and because K is an arbitrary compact subset of Ω , the limit holds for every point of Ω . Moreover, as we saw in Section 4 of Chapter 3

$$\frac{G'_N}{G_N} = \sum_{n=1}^N \frac{F'_n}{F_n},$$

so part (ii) of the proposition is also proved.

3.2 Example: the product formula for the sine function

Before proceeding with the general theory of Weierstrass products, we consider the key example of the product formula for the sine function:

$$(3) \quad \frac{\sin \pi z}{\pi} = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

This identity will in turn be derived from the sum formula for the cotangent function ($\cot \pi z = \cos \pi z / \sin \pi z$):

$$(4) \quad \pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}.$$

The first formula holds for all complex numbers z , and the second whenever z is not an integer. The sum $\sum_{n=-\infty}^{\infty} 1/(z+n)$ needs to be properly understood, because the separate halves corresponding to positive and negative values of n do not converge. Only when interpreted symmetrically, as $\lim_{N \rightarrow \infty} \sum_{|n| \leq N} 1/(z+n)$, does the cancellation of terms lead to a convergent series as in (4) above.